

Chapter 24 Back Matter Check

Contents

24 Historical Introduction	1
Problems	1
References	1

24 Historical Introduction

Problems

1. Show that the most general symmetry algebra allowed under the assumptions of the Coleman–Mandula theorem in the case where all particles are massless consists of internal symmetry generators plus either the Poincaré algebra or the conformal algebra $(??)$, $(??)$.
2. Show that the Gervais–Sakita action $(??)$ is invariant under the worldsheet supersymmetry transformation $(??)$.
3. Calculate the change in the Wess–Zumino Lagrangian density $(??)$ under the space-time supersymmetry transformation $(??)$.

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